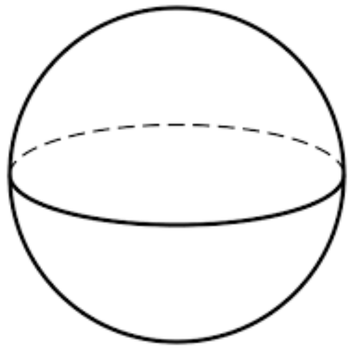
Future directions $t \to \infty$

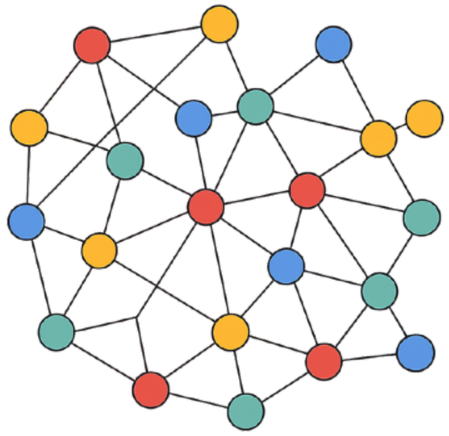
Neural networks are objects that can be fruitfully understood from principles used in physics.

Fluctuations

Symmetry

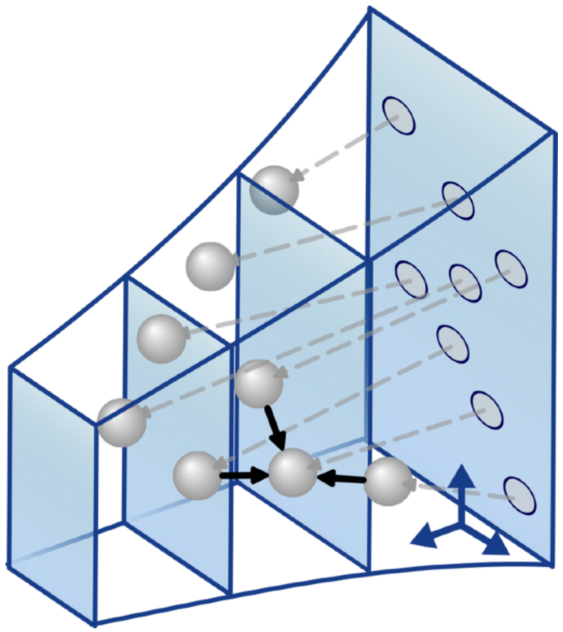


Locality



In physics these principles result in profound insights, e.g. the classification of phases of matter.

Machine learning gives us a new and fascinating class of systems to understand where **cherished physical principles** can find **new expression.** Exciting to see where these tools will take us.



The End